

Introduction to Bayesian Statistics

Part 2 Bayesian Principles

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This lecture

Frequentist statistics and Null hypothesis significance testing

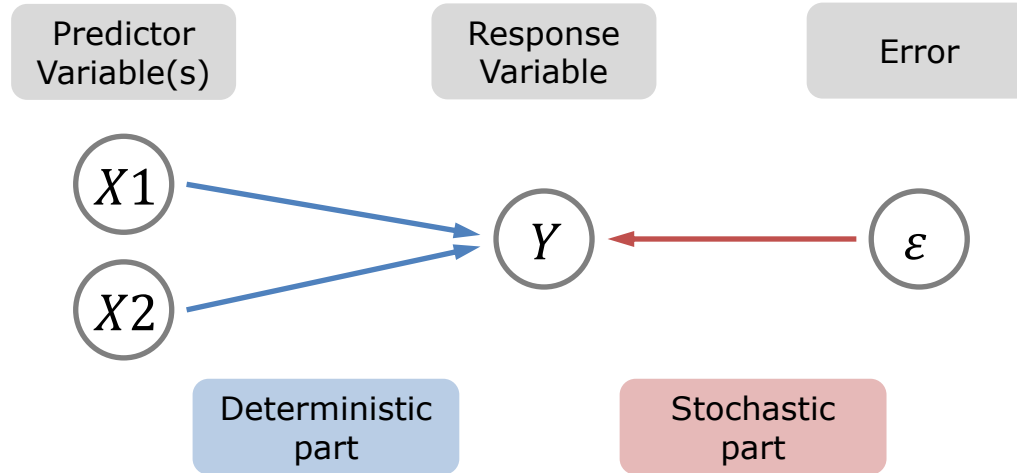
Bayes' rule

Markov Chain Monte Carlo sampling

MCMC software

Why Bayesian statistics?

Statistical modeling



Statistical modeling

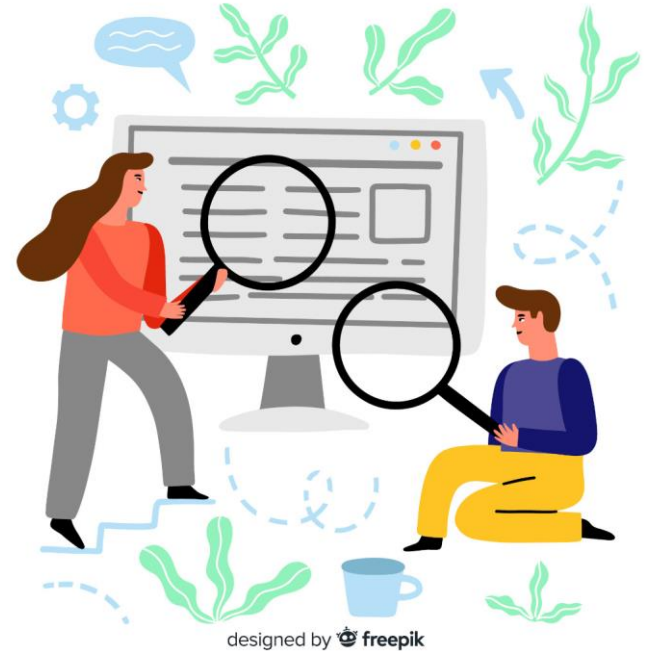
We are data detectives, trying to solve a mystery

In ecology, these mysteries can be extra tricky:

- Observational data instead of experiments
- Noisy data
- Many sources of variation

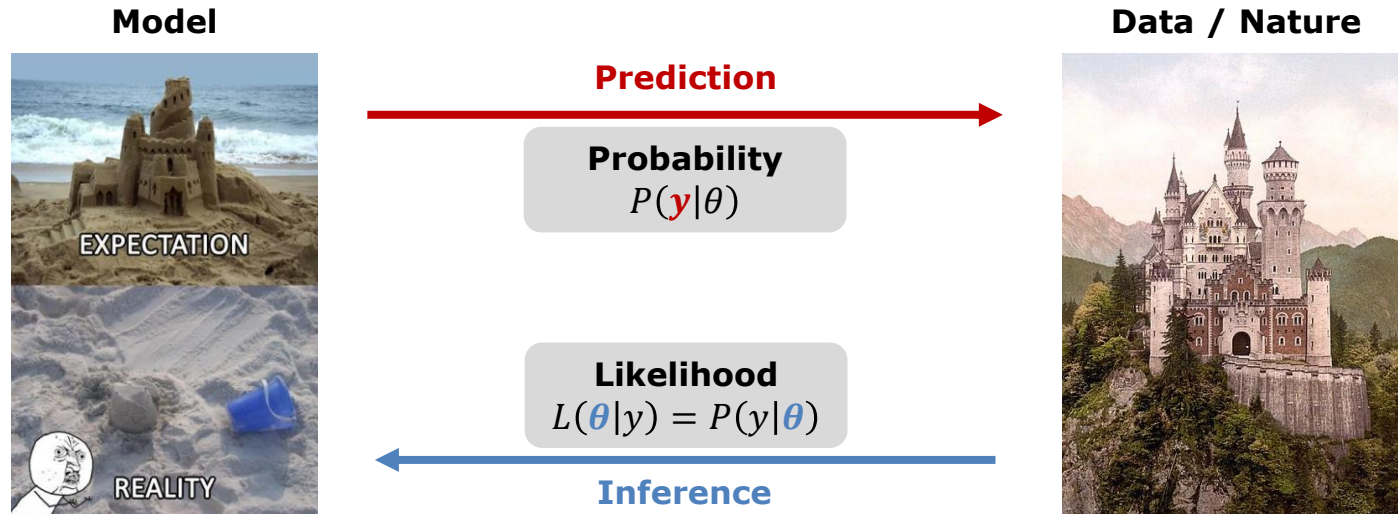
We're trying to unravel a signal from the noise
(e.g. overall trend of biodiversity loss)

We want to make quantitative statements on research questions!



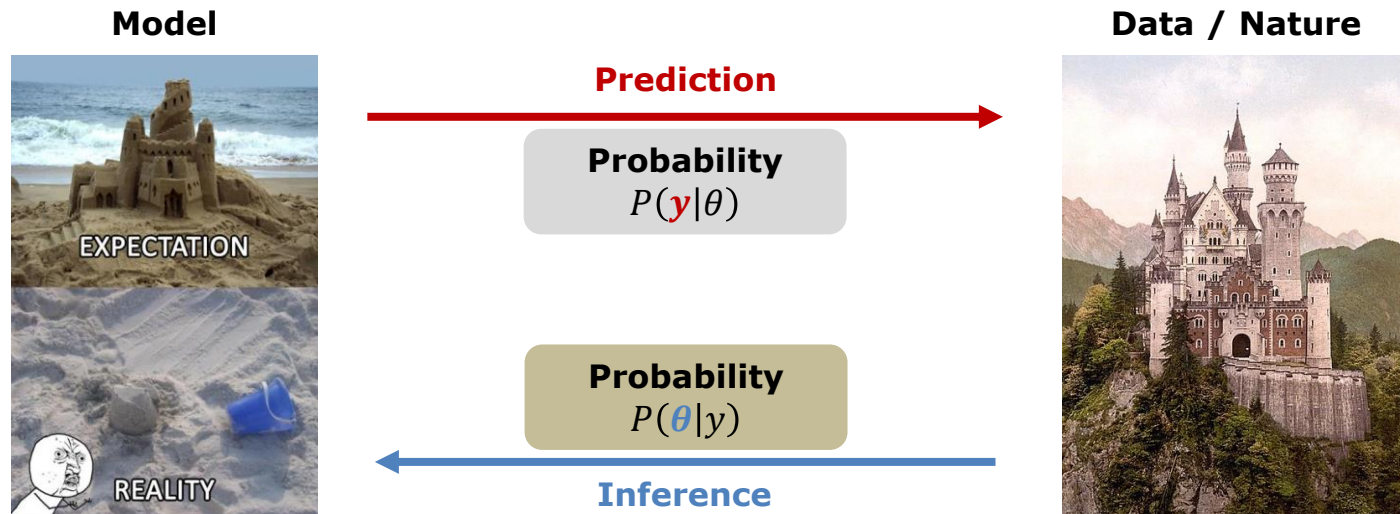
Some comments on frequentist stats

Statistical modeling



Maximum likelihood alone can't make probability statements about our research question!

Statistical modeling

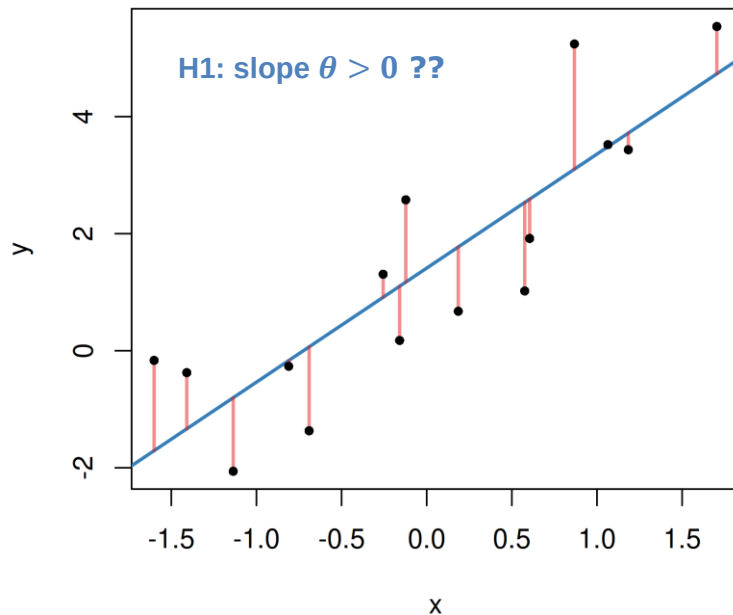


We want to assign probabilities to model parameters! E.g., $P(\theta > 0)$

The frequentist „trick”: Null hypothesis significance testing

1. We want to test hypothesis **H1**
(e.g., association is positive: $\theta > 0$)
2. We assume that the null hypothesis **H0** is true
(e.g., association is zero: $\theta = 0$)
3. Use a transformation T for which data Y (residuals) have a well-known distribution $P(T)$ under **H0**
4. If $T(Y)$ deviates enough from the assumed distribution, reject the null hypothesis
 $P(T > T(Y) \mid \theta = 0) < 0.05 \rightarrow \text{reject H0}$

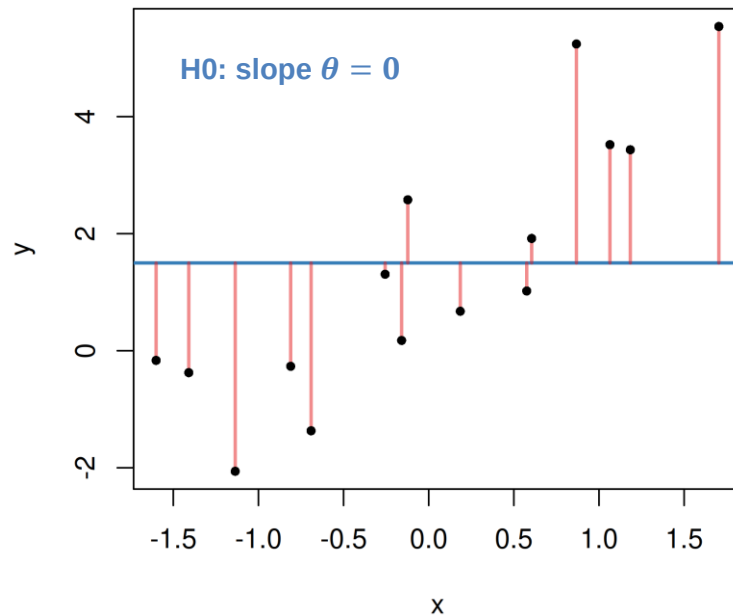
Tests if the estimated association θ^* (MLE)
is just due to randomness of the data



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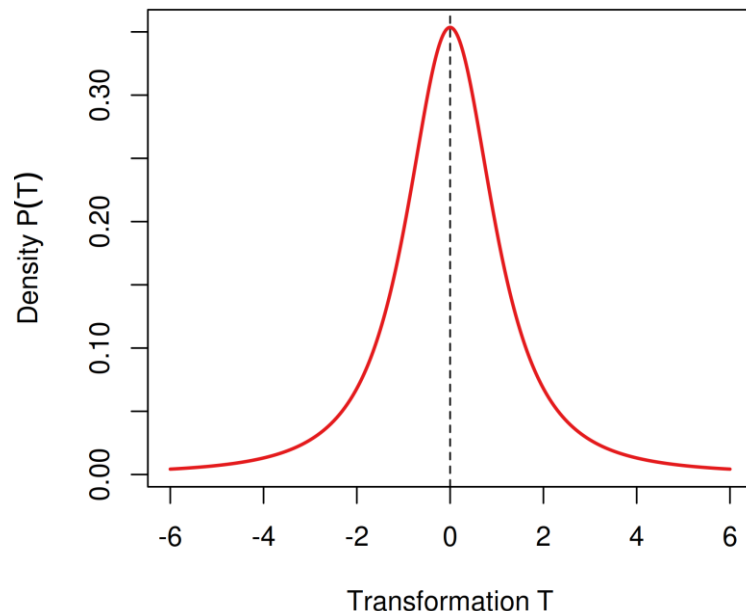
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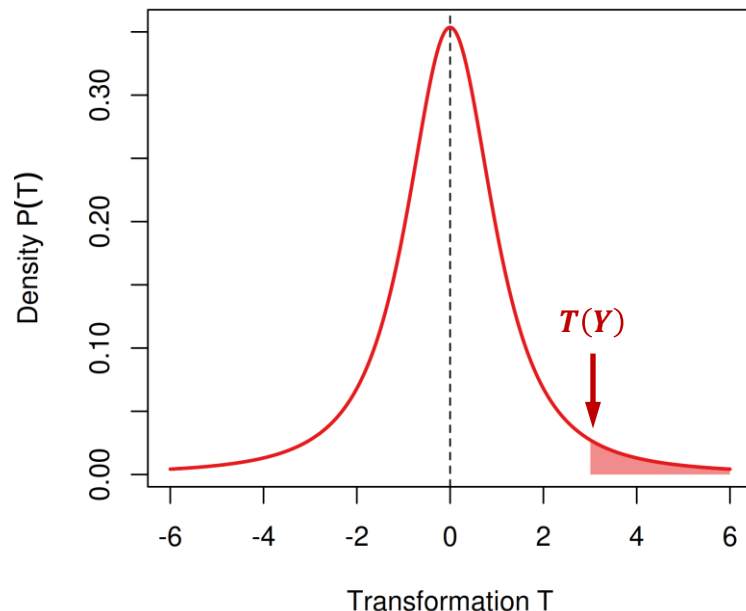
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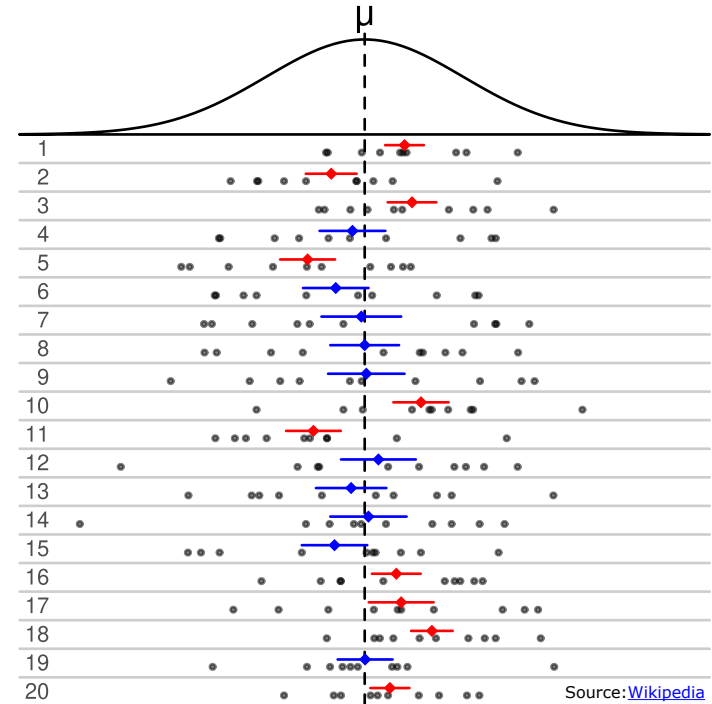


Frequentist principles

→ Data are a random realization of an experiment.
True (but unknown) parameter is fixed.

Some problems with NHST:

- P-value: Probability of the data under the null hypothesis
- Can't confirm hypotheses, just reject the null hypothesis
- Standard errors rely on assumptions & approximations
- Confidence intervals' interpretation tricky
- Limited to tests with known distributions (T-test, F-test, ...)



50%-confidence intervals in 20 repeated experiments:
10 out of 20 contain the true value μ

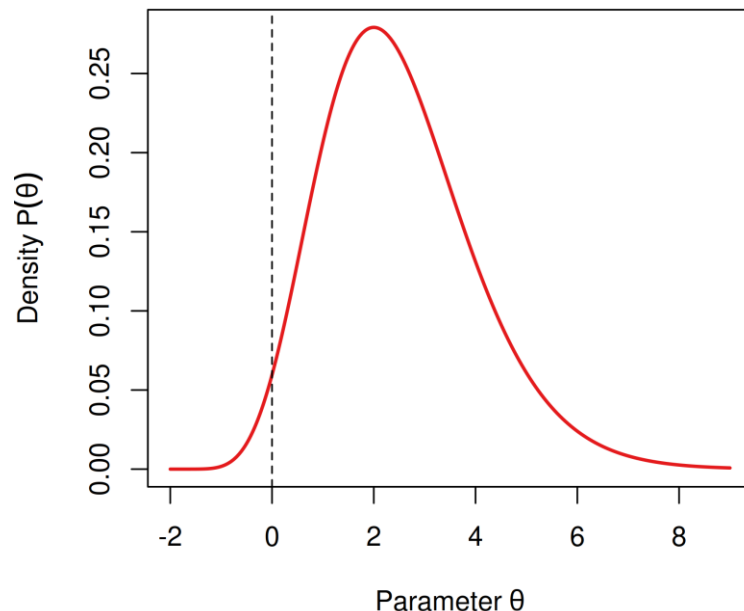
Bayesian principles

To make quantitative statements about research questions, we need probability distribution for model parameters, after observing the data $P(\theta|y)$

→ Data is fixed, parameters are random.

Some examples:

- $P(\theta > 0) = 0.99$:
"I am 99% certain that the association is positive."
- 90%-quantile [0.5, 4.3]:
"There is a 90% chance the slope is between 0.5 and 4.3."
- 2 population means μ_1 and μ_2 .
 $P(\mu_1 - \mu_2)$ quantifies distribution of population-level difference.



Bayesian models ???

There is no such thing as a „Bayesian model“!

Maximum likelihood → *Point estimates*

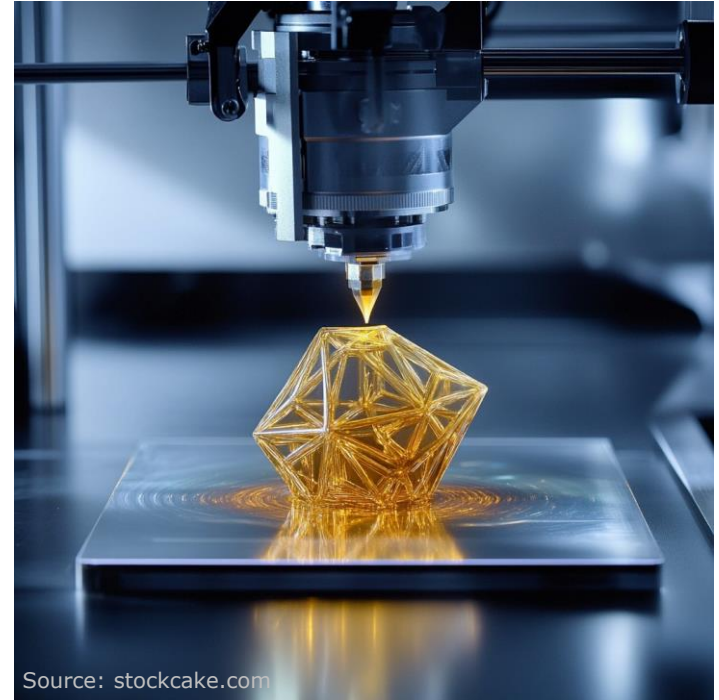
Frequentist NHST → P-values for Null hypothesis

Bayesian stats → True probability distribution
for model parameters

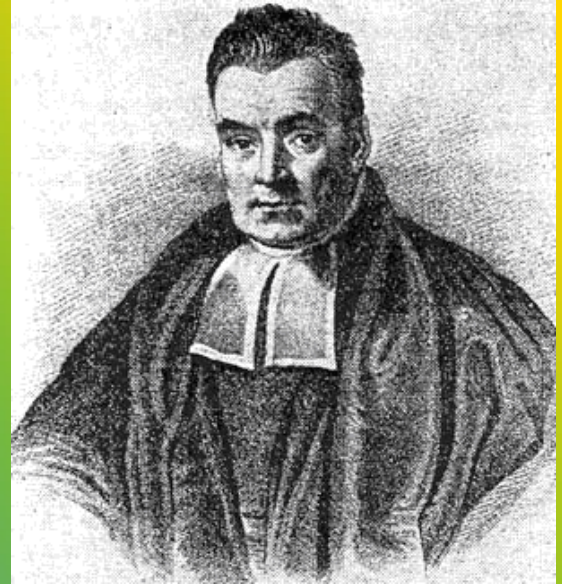
“Full luxury Bayes” (Richard McElreath)

“Bayesian 3D printer”

The Bayesian 3D printer



Bayes' rule



Reverend Thomas Bayes (1701–1761)

Bayes' rule

Conditional probability:

Prob. of event A , given that B occurred

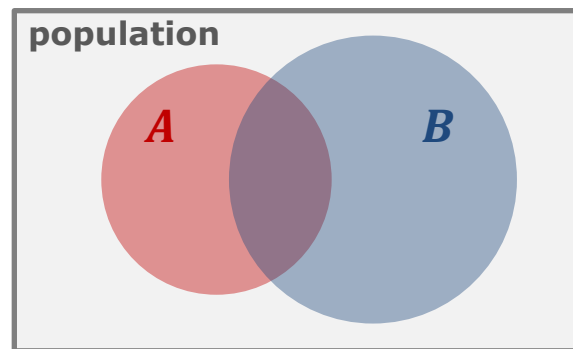
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$P(\text{Covid} | \text{positive}) = \frac{P(\text{positive}|\text{Covid}) \cdot P(\text{Covid})}{P(\text{positive})}$$

$P(\text{positive} | \text{Covid})$ test sensitivity

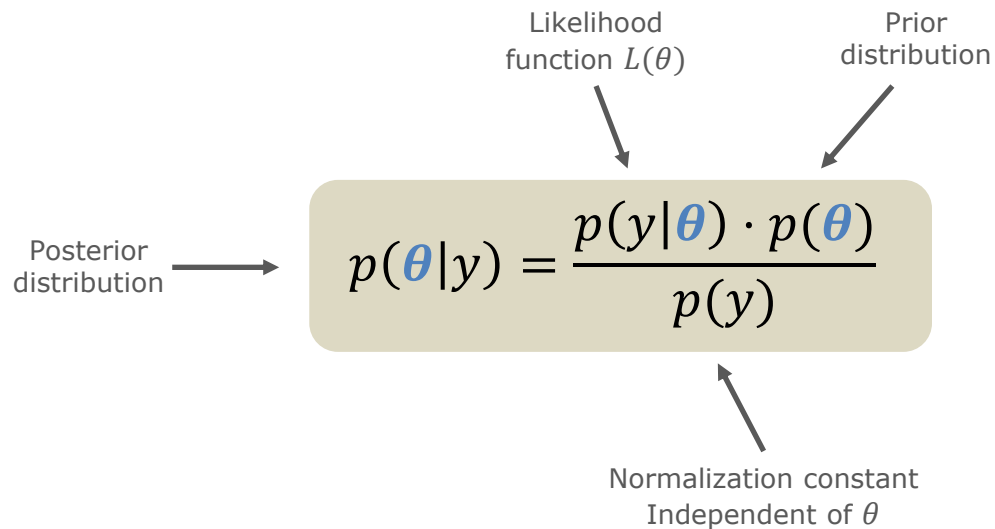
$P(\text{Covid})$ prevalence in the population

$P(\text{positive})$ positive test rate



Bayes' rule

Probability distribution of model parameters θ
after observing the data y



The diagram illustrates Bayes' rule with the formula $p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$ centered in a light beige rounded rectangle. Four arrows point to the components of the formula: 'Likelihood function $L(\theta)$ ' points to $p(y|\theta)$, 'Prior distribution' points to $p(\theta)$, 'Posterior distribution' points to $p(\theta|y)$, and 'Normalization constant Independent of θ ' points to $p(y)$.

Likelihood function $L(\theta)$

Prior distribution

Posterior distribution

$$p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$$

Normalization constant
Independent of θ

Bayes' rule

$$p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$$

$p(\theta|y)$ Posterior distribution

Update prior information in light of new evidence (data)

$p(\theta)$ Prior distribution

Belief about model parameters before data is observed

$p(y|\theta) = L(\theta)$ Likelihood function

Data inform the parameters, but L is not a probability distribution for θ

$p(y) = \int_{\theta} p(y|\theta) p(\theta)$ Normalization constant

Ensures that posterior is a true probability distribution with $\int_{\theta} p(\theta|y) = 1$

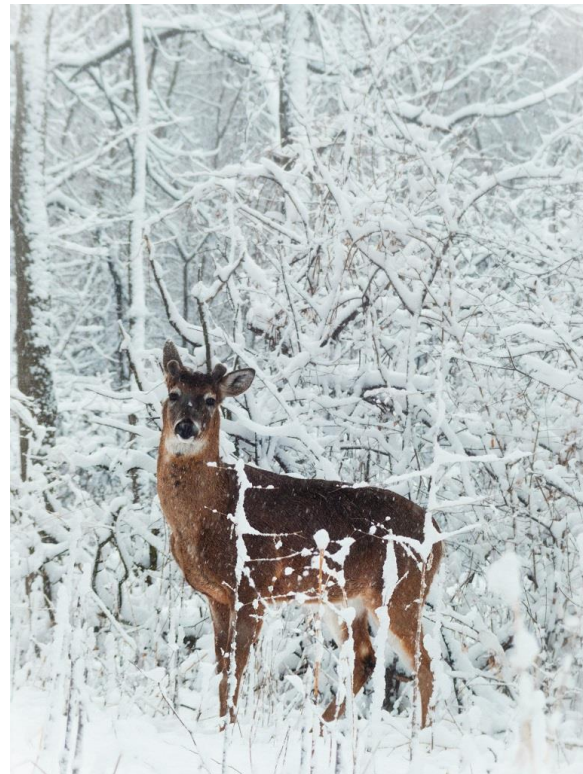
Example

Survival rate of a deer population

1 datapoint: 7 out of 10 individuals survived

Deterministic part: average survival rate θ

Stochastic part: $y \sim \text{Binomial}(N, \theta)$



Example

Survival rate of a deer population

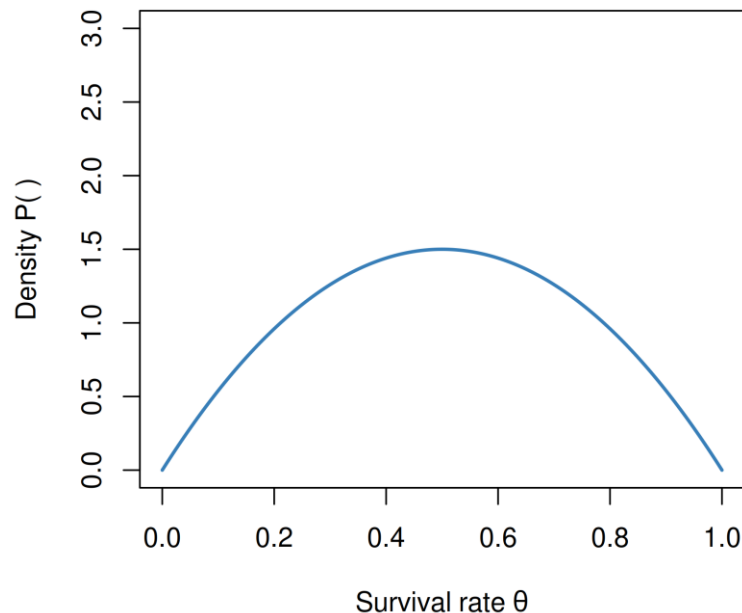
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Deterministic part: average survival rate θ

Stochastic part: $y \sim \text{Binomial}(N, \theta)$

Prior: θ almost 0 or almost 1 are improbable

$\theta \sim \text{beta}(2,2)$ or $p(\theta) = \text{dbeta}(\theta | 2,2)$



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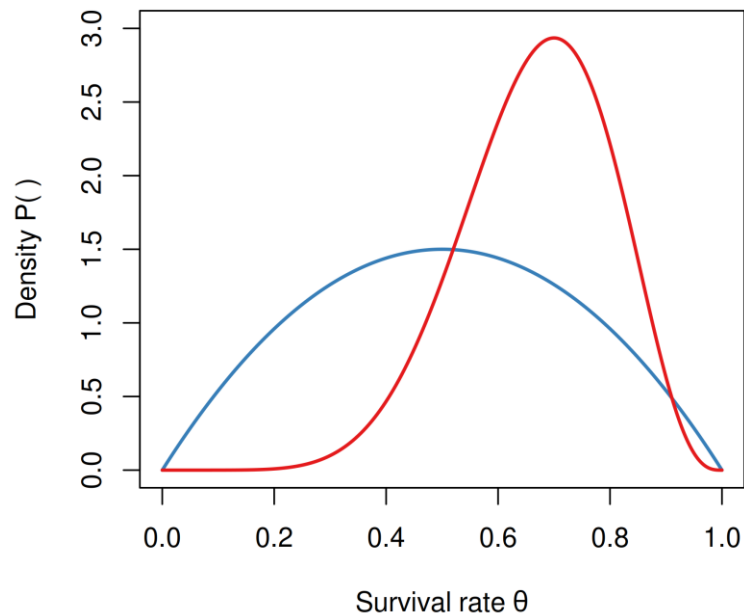
Stochastic part: $y \sim \text{Binomial}(N, \theta)$

Prior: θ almost 0 or almost 1 are improbable

$\theta \sim \text{beta}(2,2)$ or $p(\theta) = \text{dbeta}(\theta \mid 2,2)$

Likelihood: defined by statistical model & data

$L(\theta) = p(y|\theta) = \text{dBinomial}(y = 7, N = 10 \mid \theta)$



Example

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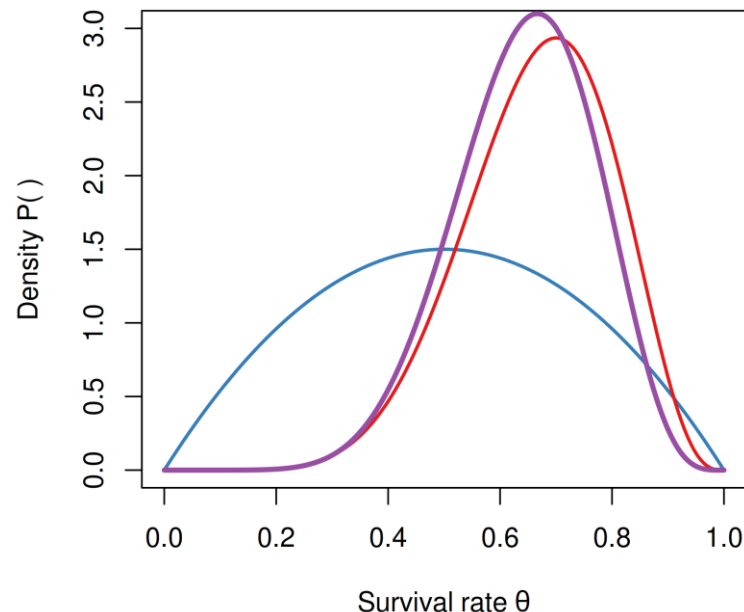
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Likelihood: defined by statistical model & data

$L(\theta) = p(y|\theta) = \text{dBinomial}(y = 7, N = 10 | \theta)$

Posterior:

$$p(\theta|y) = \frac{L(\theta) \cdot p(\theta)}{c}$$



Example: different prior

Survival rate of a deer population

1 datapoint: 7 out of 10 individuals survived

Deterministic part: average survival rate θ

Stochastic part: $y \sim \text{Binomial}(N, \theta)$

Prior: uninformative

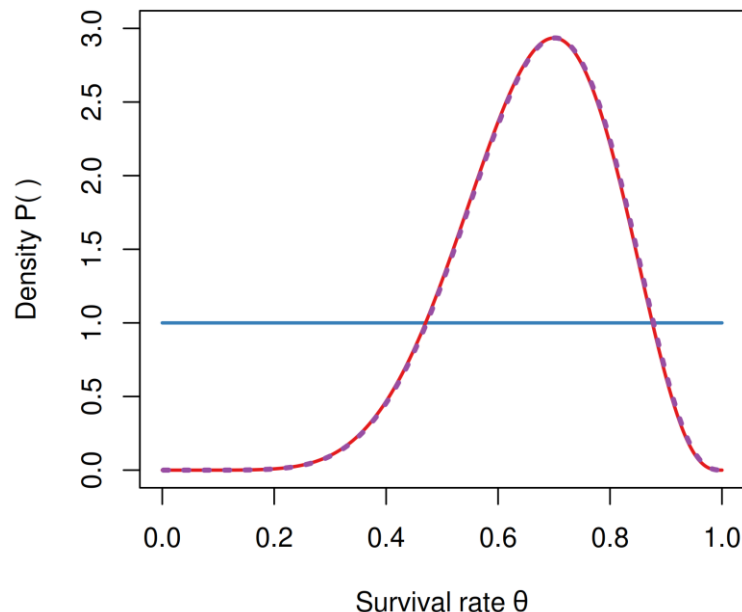
$\theta \sim \text{beta}(1,1) = \text{uniform}(1,1)$

Likelihood: defined by statistical model & data

$L(\theta) = p(y|\theta) = \text{dBinomial}(y = 7, N = 10 | \theta)$

Posterior:

$p(\theta|y) = \frac{L(\theta) \cdot p(\theta)}{c}$ proportional to likelihood here



Calculation of the posterior?

To compute, e.g. $\text{mean}(\theta)$, $\text{sd}(\theta)$, $P(\theta > 0.5)$...

we'd need to know $p(\theta|y)$ for all θ and also $c = \int_{\theta} L(\theta)p(\theta)$

1) Analytical (mathematical formula)

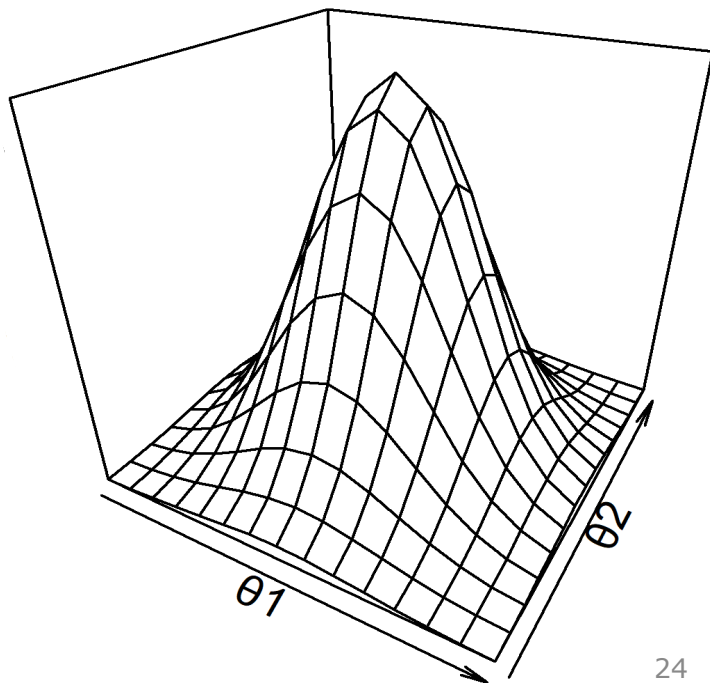
→ Much too complicated, often impossible

2) Numerical (e.g., grid)

→ Effort grows exponentially with $\#\theta$

→ Computationally too expensive

„Curse of
dimensionality“



Oh no! Same problem as before.

Markov Chain Monte Carlo (MCMC) sampling

New idea: sampling!

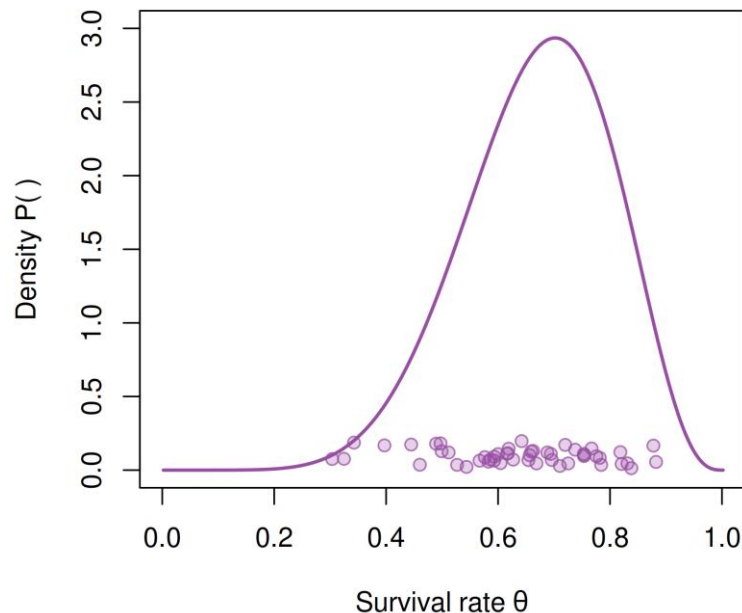
$$p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$$

Instead of calculating $p(\theta|y)$, draw random θ samples.
Many samples where p high, few samples where p low

- Sample density proportional to $p(\theta|y)$
- We don't need the normalizing constant $c = p(y)$

$$p(\theta|y) \sim p(y|\theta) \cdot p(\theta)$$

Posterior is proportional to likelihood x prior



Markov Chain Monte Carlo

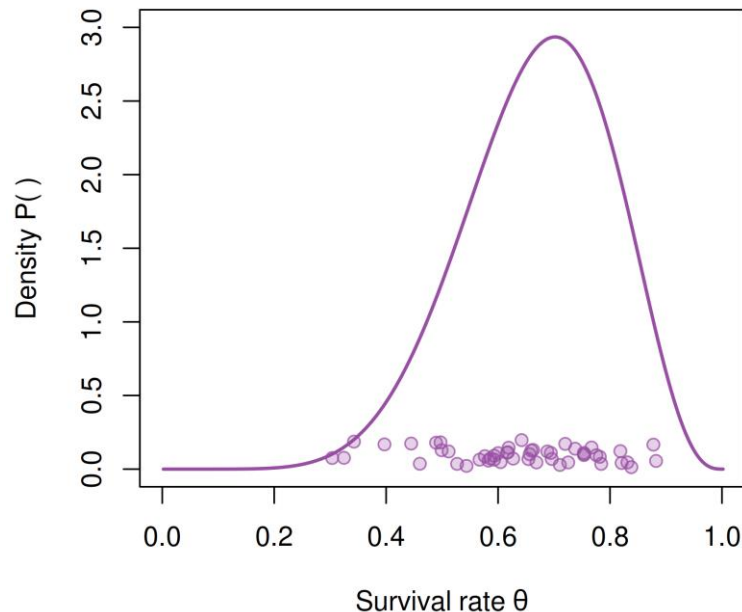
Start with initial θ_1

Compute $f(\theta_1) = L(\theta_1) \cdot p(\theta_1)$

In each step $i = 2, 3, 4, \dots$:

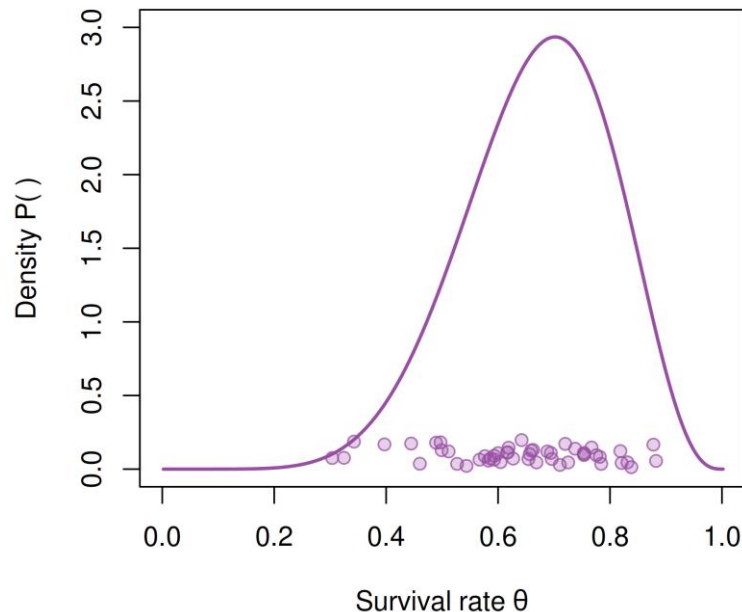
- Propose new θ_{new} , e.g. $\theta_{\text{new}} \sim \text{Normal}(\theta_{\text{old}}, \sigma)$
Compute $f(\theta_{\text{new}}) = L(\theta_{\text{new}}) \cdot p(\theta_{\text{new}})$
- If $f(\theta_{\text{new}}) > f(\theta_{\text{old}})$
→ accept $\theta_{i+1} = \theta_{\text{new}}$
- If $f(\theta_{\text{new}}) < f(\theta_{\text{old}})$
→ accept $\theta_{i+1} = \theta_{\text{new}}$ with probability $\frac{f(\theta_{\text{new}})}{f(\theta_{\text{old}})}$
(random draw)
→ otherwise reject θ_{new}

Repeat e.g. 1000 times



Markov Chain Monte Carlo

- $\theta_1, \theta_2, \theta_3, \dots, \theta_{1000}$ are called the „**chain**“
- They are samples from the underlying (but mathematically unknown) posterior distribution
- We can calculate empirical quantities
 - mean
 - standard deviation
 - quantiles
 - histogram (for visualization)
 - probability statements
- Without explicitly knowing the function $p(\theta|y)$
- Even don't need to save computed values $p(\theta_1|y), p(\theta_2|y), \dots$



Markov Chain Monte Carlo

„**Markov**“ property:

each sample θ_i **only** depends on the previous sample θ_{i-1}

„**Chain**“:

list of samples $\theta_1, \theta_2, \theta_3, \dots, \theta_{1000}$

„**Monte Carlo**“:

each new sample involves a random draw

Very simple algorithm at its core (few lines of code)

Very sophisticated software to make it efficient
(lots of maths go into good sample proposals)



Example

Survival rate: 1 datapoint (7/10), prior $\text{beta}(2,2)$

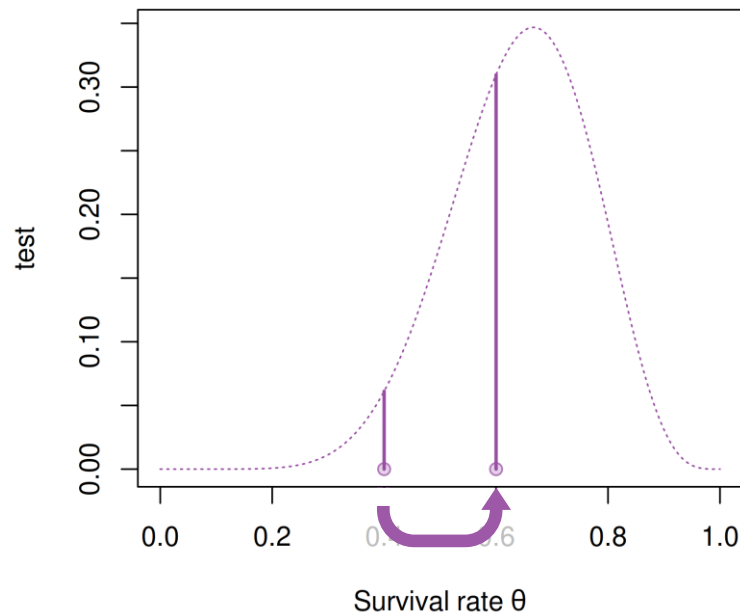
First sample $\theta_1 = 0.4$

$$\begin{aligned} f(0.4) &= \text{dbinom}(7,10,0.4) \cdot \text{dbeta}(0.4,2,2) \\ &= 0.061 \end{aligned}$$

Propose $\theta_{\text{new}} = 0.6$

$$\begin{aligned} f(0.6) &= \text{dbinom}(7,10,0.6) \cdot \text{dbeta}(0.6,2,2) \\ &= 0.310 \end{aligned}$$

$f(\theta_{\text{new}}) > f(\theta_{\text{old}}) \rightarrow$ accept $\theta_2 = 0.6$ as next sample



Example

Survival rate: 1 datapoint (7/10), prior beta(2,2)

Current sample $\theta_2 = 0.6$

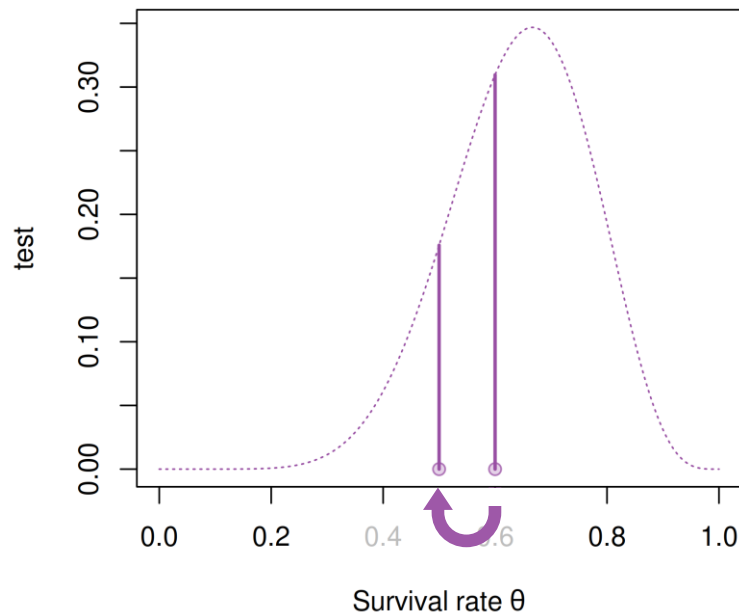
$$f(0.6) = 0.310$$

Propose $\theta_{\text{new}} = 0.5$

$$\begin{aligned} f(0.5) &= \text{dbinom}(7,10,0.5) \cdot \text{dbeta}(0.5,2,2) \\ &= 0.175 \end{aligned}$$

$$f(\theta_{\text{new}}) < f(\theta_{\text{old}}) \rightarrow \text{accept with probability } \frac{f(\theta_{\text{new}})}{f(\theta_{\text{old}})} = \frac{0.175}{0.310} = 0.564$$

→ random draw with a 56.4% chance
to accept $\theta_3 = 0.5$ as next sample



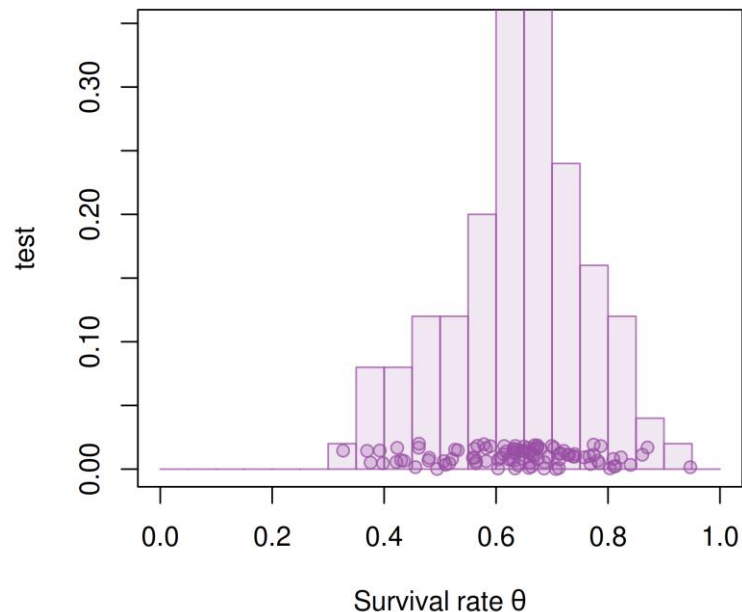
Example

Survival rate: 1 datapoint (7/10), prior $\text{beta}(2,2)$

Samples from posterior distribution $\theta_1, \theta_2, \theta_3, \dots$
describe probability for estimated survival rate.

Don't need to know the full curve $p(\theta|y)$!

- Empirical **histogram** for visualization



Example

Survival rate: 1 datapoint (7/10), prior $\text{beta}(2,2)$

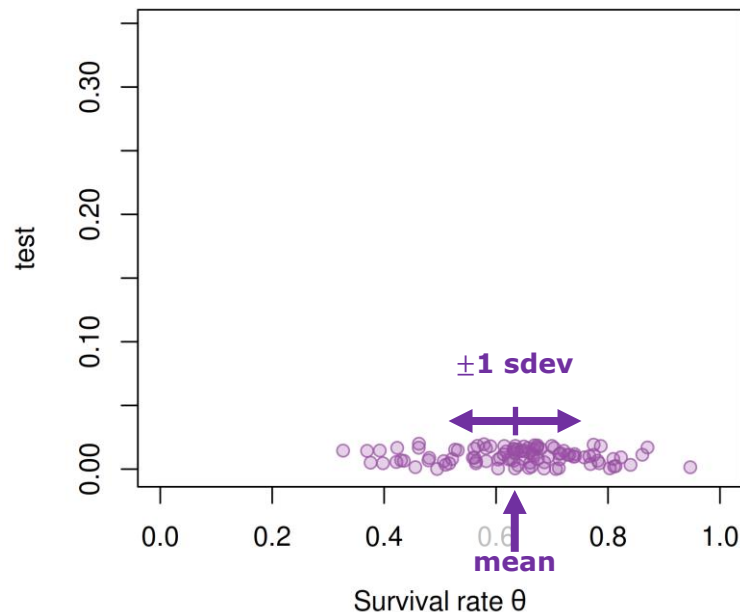
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describe probability for estimated survival rate.

Don't need to know the full curve $p(\theta|y)$!

- Empirical **mean** and **standard deviation**

$$\text{mean} = \frac{1}{K} \sum_{i=1}^K \theta_i$$

$$\text{sdev} = \sqrt{\frac{1}{K} \sum_{i=1}^K (\theta_i - \text{mean})^2}$$



Example

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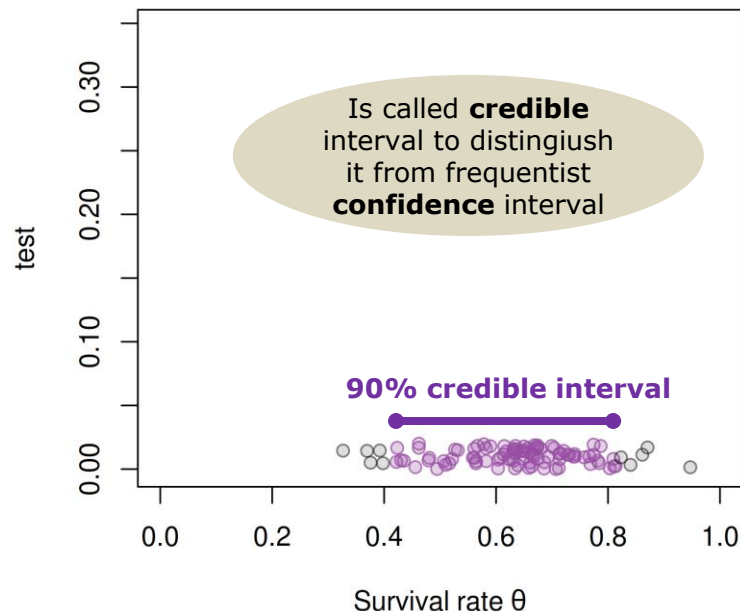
Don't need to know the full curve $p(\theta|y)$!

- „Credible intervals“

90% of samples between the 5% and the 95% quantiles.

$$P(\theta \in [0.42, 0.81]) = 0.9$$

„I am 90% sure the survival probability is between 0.42 and 0.81“



Example

Survival rate: 1 datapoint (7/10), prior $\text{beta}(2,2)$

Samples from posterior distribution $\theta_1, \theta_2, \theta_3, \dots$
describe probability for estimated survival rate.

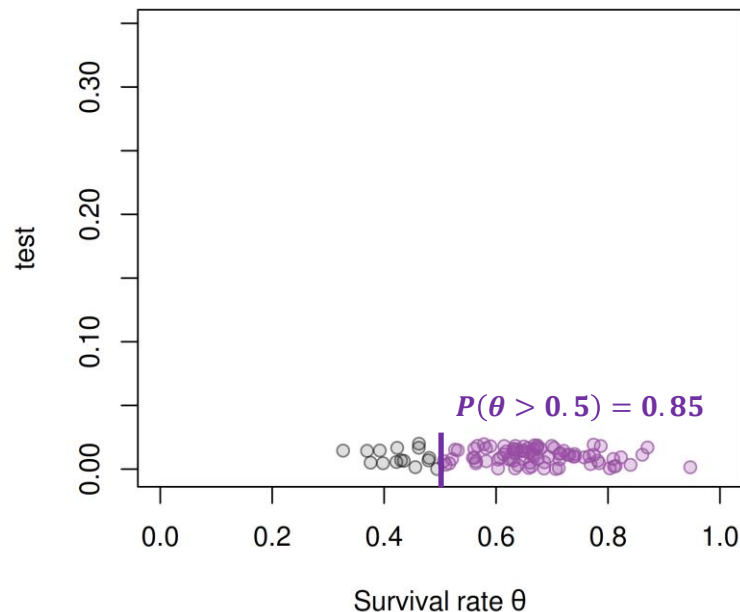
Don't need to know the full curve $p(\theta|y)$!

- **Probability statements** (about hypotheses)

85% of samples larger than a survival rate of 0.5

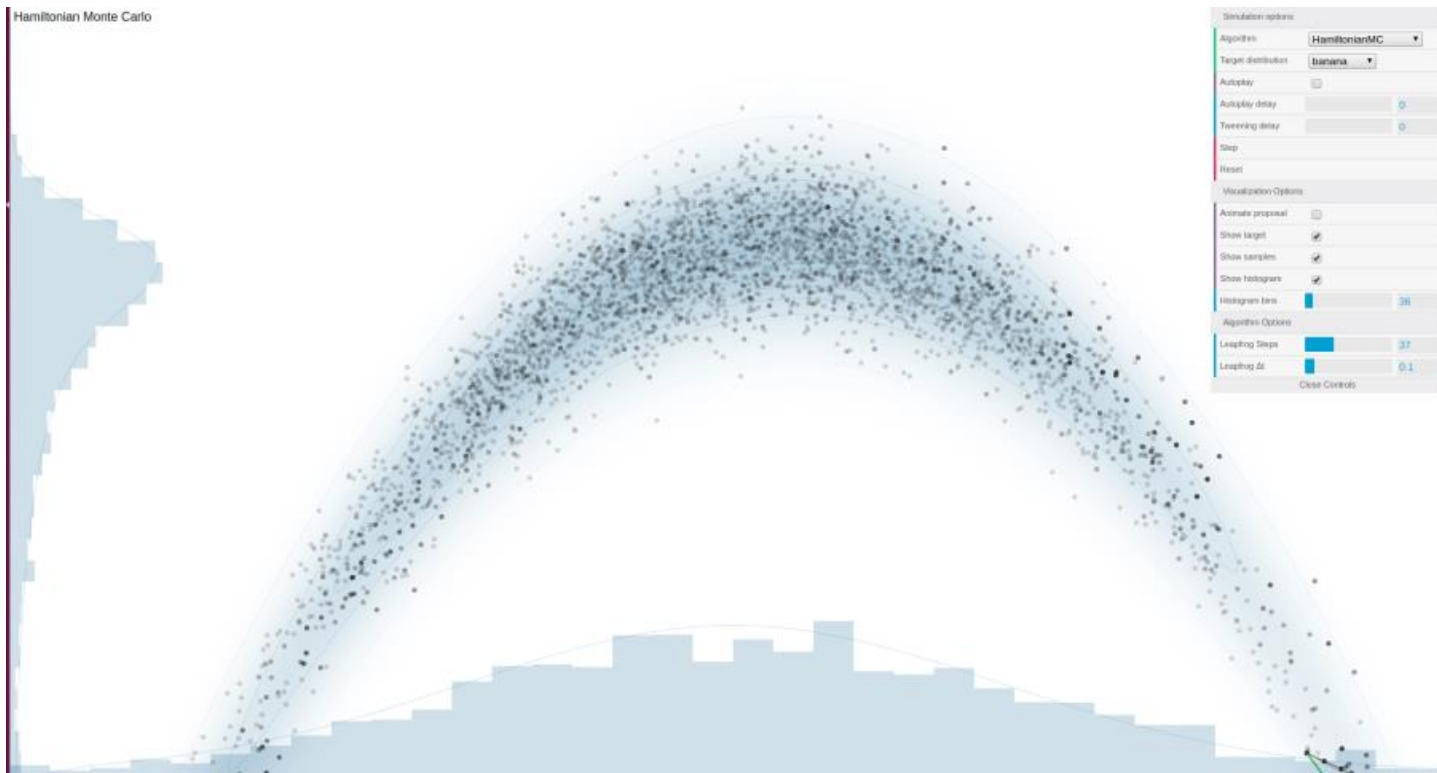
$$P(\theta > 0.5) = 0.85$$

„I am 85% sure the survival probability is larger than 0.5“



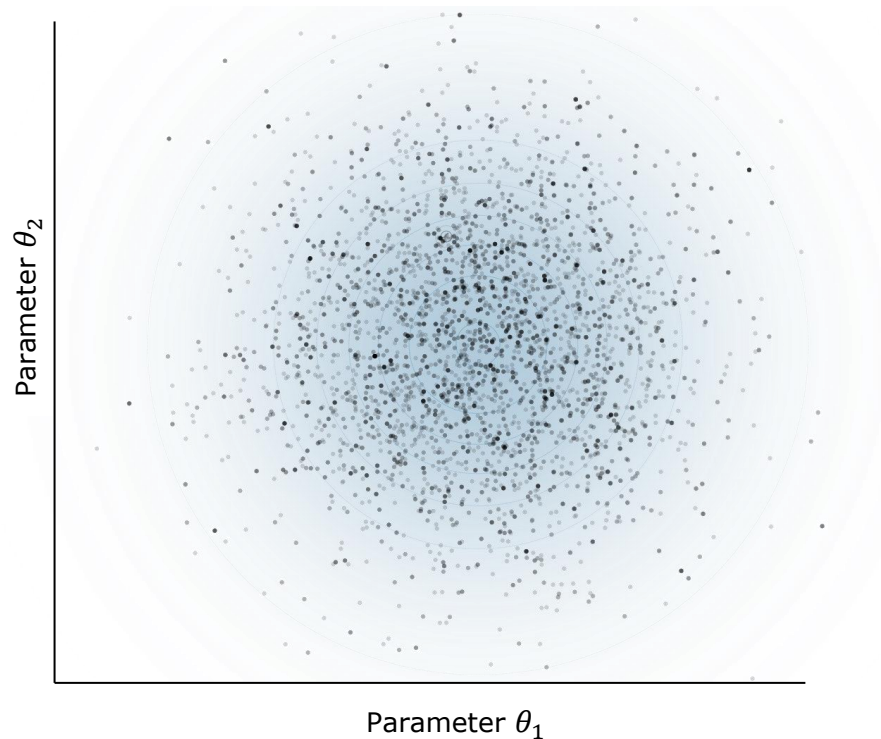
MCMC Demo

<https://chi-feng.github.io/mcmc-demo/app.html>



Convergence

- Mathematical theory says that MCMC will *eventually* be a good approximation of the posterior distribution
- How many samples are enough?
- Start with 1000-2000 samples
- Run multiple chains (3-4)
- Visual inspection
- Quantitative measures

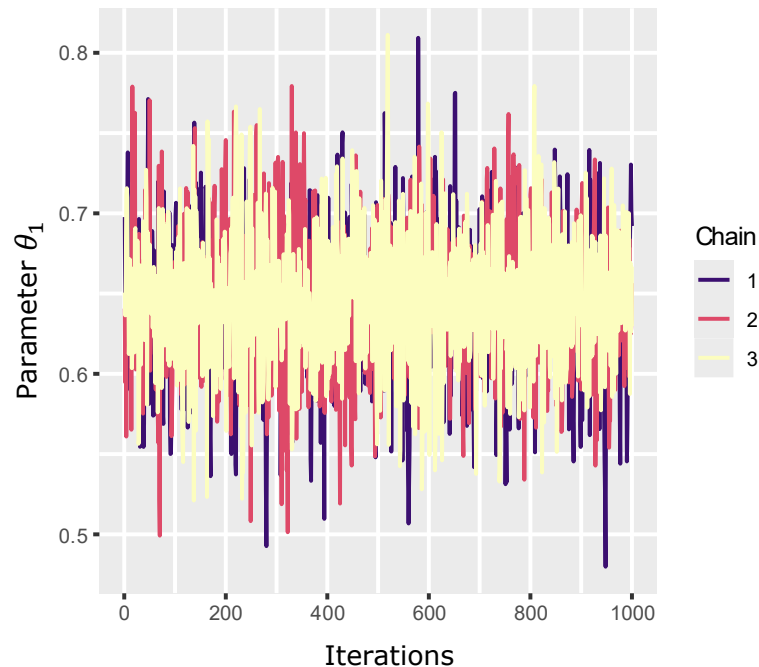


Convergence

Visual inspection

- Traceplots for each parameter
- Should look like random noise
- Centered around a constant mean
- Chains should look similar
- Like a fuzzy caterpillar!

→ MCMC has converged

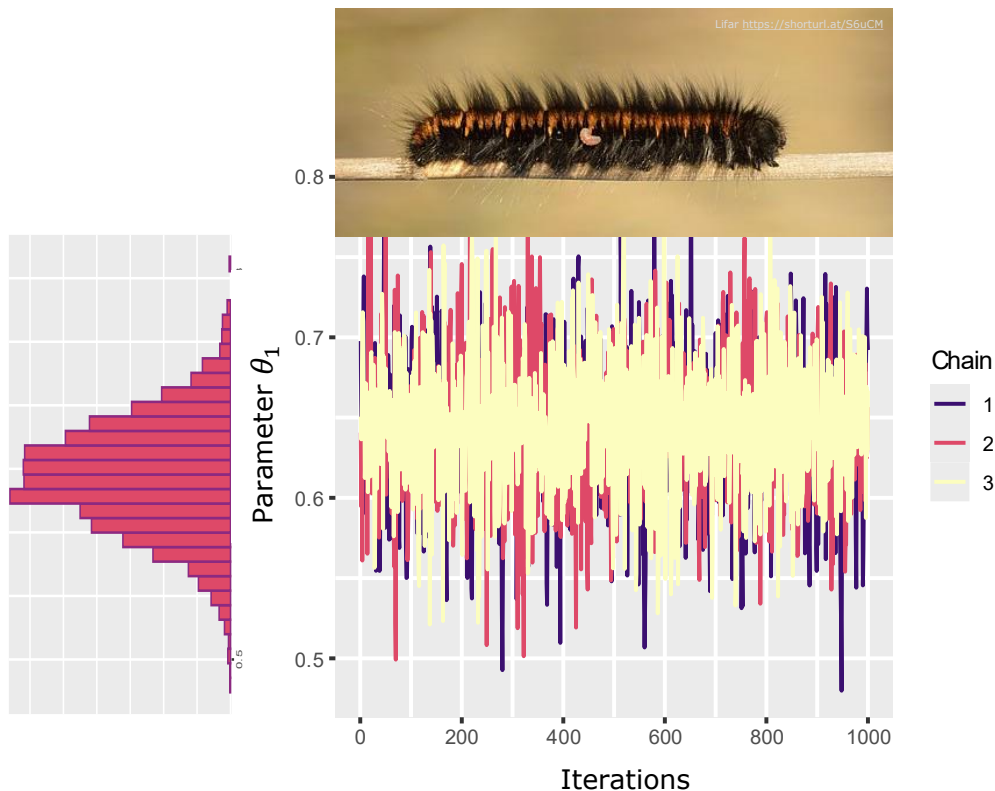


Convergence

Visual inspection

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→ **MCMC has converged**



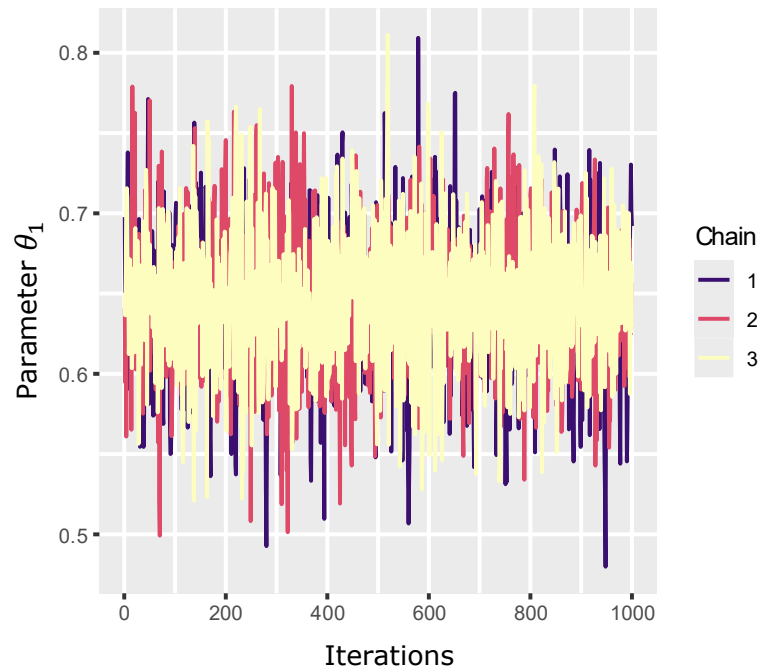
Convergence

Quantitative measures

- **Rhat** value („Gelman-Rubin statistic")
 - Compares the variation within and across chains
 - Value should be less than **1.1**
- **n_eff** (Number of effective samples)
 - Chains usually have a bit of autocorrelation but it shouldn't be too strong
 - Small n_eff values indicate a problem

→ MCMC has converged

```
3 chains with 1000 samples each
3000 samples (post-warmup)
Rhat = 1.001
n_eff= 1770
```



Software

Some history



1700s Bayes' theorem, Laplace formalized it

Bayes impractical
Restricted to simple cases

Early 1800s Gauß: least squares, regression

Late 1800s to early 1900s

Birth of modern statistics. Pearson, Fisher, Neyman ... :
max. likelihood, hypothesis testing, design of experiments

Frequentism superseded Bayes
More practical in most cases

Mid to late 1900s MCMC algorithms

Still a niche topic in statistics

2000s Computational tools for MCMC
BUGS, JAGS, Stan ...

Becoming more popular in sciences

Today Convenient R interfaces
brms, rstanarm ...

Taught in gradschools

Future

Becoming the default
instead of frequentism ??

Software

All Bayesian software contains:

1) Modeling language

User must define statistical model:

parameters, likelihood & priors

2) MCMC sampler

Automated algorithm that takes care of sampling

Bayesian programming languages

- + Maximum flexibility in statistical modeling
- + Total control over every part of the model
- Steep learning curve
- Coding can be time-consuming

JAGS

Was the most popular once, now less used

Nimble



Extends JAGS, more flexible

Stan



Very efficient, runs in C++

Can all
be called
from **R**

```
a ~ dnorm(0, sd=1)
b ~ dnorm(0, sd=1)
sigma ~ dexp(1.0)
for(i in 1:n) {
  y[i] ~ dnorm(a+b*x[i], sd=sigma)
}
```

Nimble

```
data {
  int<lower=0> N;
  vector[N] x;
  vector[N] y;
}
parameters {
  real a;
  real b;
  real<lower=0> sigma;
}
model {
  a ~ normal(0,1);
  b ~ normal(0,1);
  sigma ~ exponential(1.0);
  y ~ normal(a+b*x, sigma);
}
```

Stan

Formula-based R packages

- + Model formulation similar to `lm` or `lme4`
- + Easy to learn
- + Less coding necessary
- + Handy functions for model analysis (after fitting)
- Limited to pre-defined model types
- „lm“-formulas deceive you into forgetting about model definition

rstanarm

GLMMs only

brms

Much more flexible, becoming quite popular

Both
automatically
translate model
into **Stan**

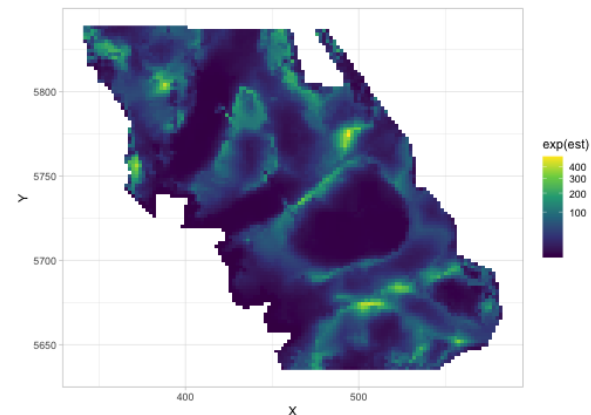


```
> stan_glm(y ~ x,  
           prior = normal(0,1),  
           data = dataset)
```

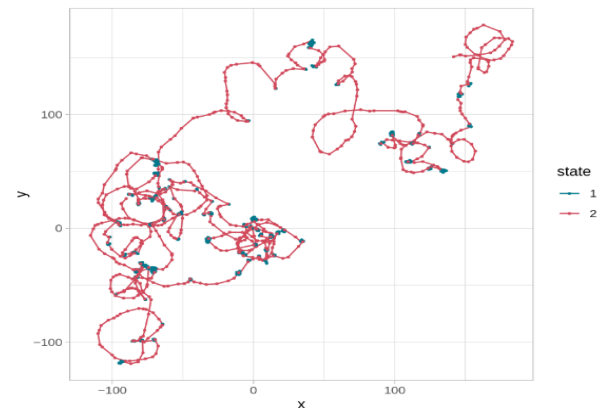
```
> brm(y ~ x,  
      prior = prior(normal(0,1),coef=x),  
      data = dataset)
```

Specialized software (R-packages)

- R-INLA, sdmTMB, → Spatial models
spBayes
- bsam, hmmTMB, → Animal movement
bayesmove
- spOccupancy → Occupancy models
- spAbundance → Abundance models
- blavaan → Structural equation models
- ...



<https://pbs-assess.github.io/sdmTMB/>



<https://github.com/TheoMichelot/hmmTMB>

Why Bayesian ?

Bayesian workflow

- 1) Research question (hypotheses)
 - 2) Data collection
 - 3) Statistical model
 - 4) Prior distribution choice
 - 5) Model fitting (MCMC)
 - 6) Evaluate model output
 - 7) Quantitative statements on hypotheses
- Revise model



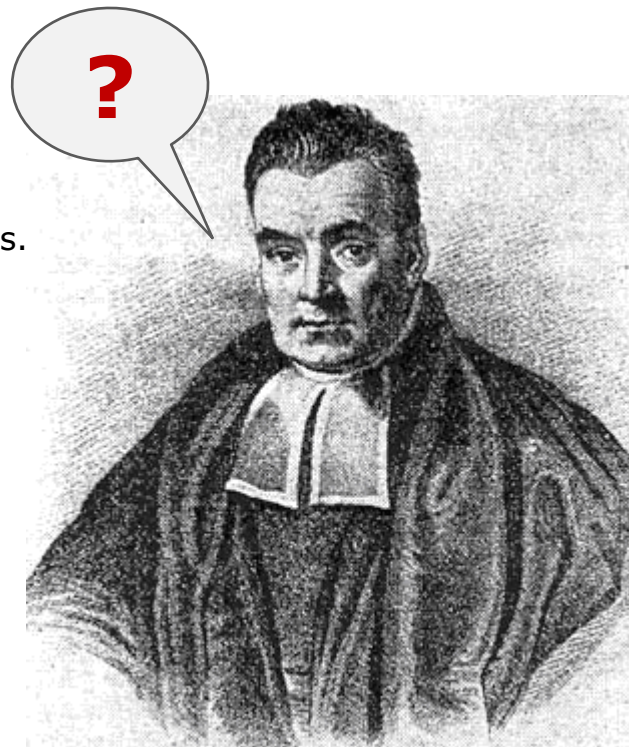
→ Workflow not that different from frequentist statistic

designed by  freepik.com

Why Bayesian?

Philosophical answer:

- Frequentism assumes true and fixed underlying parameter values.
- Data are just a sample of the „true“ statistical model.
- Bayesian statistics embraces uncertainty and wants to quantify it correctly.
- Observed data are given, model parameters uncertain.



Why Bayesian?

Practical answer:

- Output is more intuitive:
Direct inference on parameters / hypotheses instead of NHST
What does the data tell me about my model?
- Full transparency and control over model and output
- Include prior belief / information
- Parameter regularization may be necessary
- Not limited to a specific toolbox, but full flexibility in modeling
(especially with Stan or Nimble, but brms also very versatile)
- Fit complex models with lots of parameters

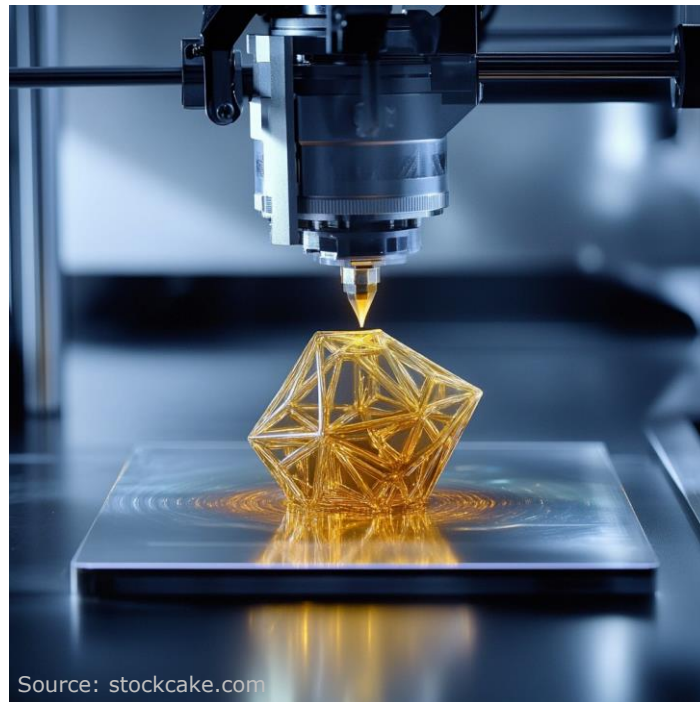


Why Bayesian?

What are complex models?

- Nonlinear models
- Hierarchical structure, mixed effects
- Combination of multiple, heterogeneous data sources and/or models
- Constraints on parameters
- Latent variables
(occupancy models, animal movement models, SEM, HMM, ...)

The Bayesian 3D printer



Source: stockcake.com

Summary

- There is no such thing as a „Bayesian model“!
- Frequentist and Bayesian stats are different methodologies for estimating parameters of a statistical model
- **Frequentist** statistics cannot (mathematically) do direct inference $P(\theta|y)$, and requires a (methodological) detour via **NHST** $P(y|\theta = 0)$
- **Bayesian** statistics can (conceptually) do direct inference $P(\theta|y)$, but requires a (computational) detour via **MCMC**

Further reading

Bürkner, P. (2024). The brms Book [in progress]. <https://paulbuerkner.com/software/brms-book/>

Fieberg, J. (2024). Statistics 4 Ecologists. <https://statistics4ecologists-v2.netlify.app/> [Chapters 11-13]

Inchausti, P. (2023). Statistical Modeling With R: a dual frequentist and Bayesian approach for life scientists. *Oxford University Press*. [Chapter 3]

Kery, M. & Kellner, F. (2024): Applied Statistical Modelling for Ecologists. *Elsevier*. [Chapter 2]

Johnson, A. A., Ott, M. Q., Dogucu, M. (2021). Bayes Rules! *CRC Press*. <https://www.bayesrulesbook.com/> [Chapters 1-2]

McElreath, R. (2020). Statistical Rethinking: A Bayesian Course with Examples in R and STAN (2nd ed.). *Chapman and Hall/CRC*. <https://doi.org/10.1201/9780429029608>

van de Schoot, R., Depaoli, S., King, R., et al. (2021). Bayesian statistics and modelling. *Nature Reviews. Methods Primers*, 1(1), 1–26. <https://doi.org/10.1038/s43586-020-00001-2>